EXPERMENT: 7

PERFORMANCE EVALUATION OF AM GENERATION AND

DETECTION UNDER NOISY CHANNEL CONDITIONS USING

MATLAB

AIM:

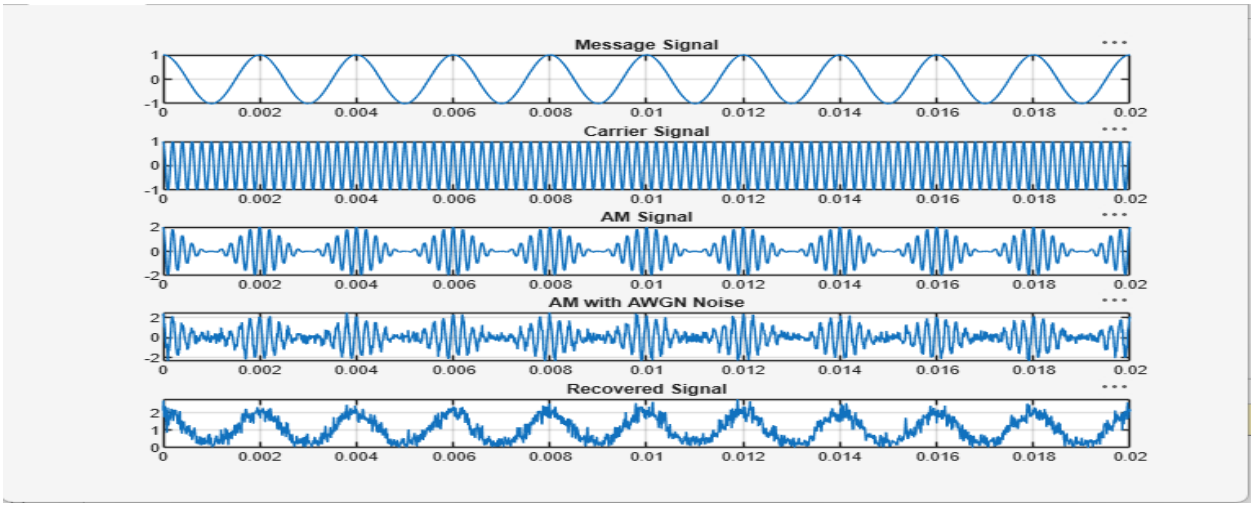
i) To generate an Amplitude Modulated (AM) signal, introduce Additive White Gaussian Noise (AWGN), and analyze the effect of noise on the transmitted signal using MATLAB.

ii) To recover the original message signal from the noisy AM signal using envelope detection and compare the transmitted and recovered waveforms to evaluate the performance of the AM communication system.

CODE:

```
clc; clear; close all;
fs=100000; t=0:1/fs:0.02;
fm=500; fc=5000;
m=cos(2*pi*fm*t);
c=cos(2*pi*fc*t);
am=(1+m).*c;
noisy=awgn(am,10,'measured');
env=abs(hilbert(noisy));
figure;
subplot(5,1,1); plot(t,m); title('Message'); grid on;
subplot(5,1,2); plot(t,c); title('Carrier'); grid on;
subplot(5,1,3); plot(t,am); title('AM Signal'); grid on;
subplot(5,1,4); plot(t,noisy); title('AM with AWGN'); grid on;
subplot(5,1,5); plot(t,env); title('Recovered Signal'); grid on;
```

OUTPUT:



RESULT:

i) The AM signal was successfully generated in MATLAB, and the effect of Additive White Gaussian Noise (AWGN) on the transmitted signal was observed through the noisy AM waveform.

ii) The original message signal was successfully recovered using envelope detection, demonstrating the effectiveness of the AM receiver in recovering the transmitted information in the presence of AWGN noise.